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A REVIEW PAPER ON SUPPLY CHAIN MANAGEMENT USING BLOCKCHAIN

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ABSTRACT

Supply Chain Management is a complex and global process that involves a wide range of stakeholders, including manufacturers, suppliers, distributors, retailers, and consumers. These stakeholders need to be able to share information about the movement of goods and data in a secure and transparent way. However, traditional SCM methods are often plagued by data silos, lack of transparency, and fraud. Blockchain technology can address these challenges by providing a shared, immutable record of all transactions that occur within a supply chain. This record is tamper-proof and accessible to all authorized participants, which can help to improve transparency and trust. Blockchain can also automate many of the manual processes involved in SCM, which can help to reduce costs and improve efficiency. This project aims to explore the potential of blockchain technology in revolutionizing supply chain management (SCM) practices. Blockchain's inherent characteristics of decentralization, immutability, and security offer promising solutions to address the prevalent challenges faced by modern supply chains, including lack of transparency, data silos, and inefficiencies.

Keywords: Supply chain management, Block chain, Blue print, Cost Benefit Analysis.

PLANNING OF THE PROJECT

Stage	Period
Research	3 weeks
Feasibility Study	1-2 weeks
System Design	2-3 weeks
Proof of concept Development	1 month
Blockchain platform selection	2-3 weeks
Integration	1 month
User Interface Design	1 month
Business Process Redesign	2 months
Testing	1 month
Pilot Program Design	2 weeks
Execution	1 month

theoretical; it is already being realized in various industries:

- Food Supply Chain: Blockchain is used to track food products from farm to table, ensuring
 - Food safety and authenticity. Consumers can scan a QR code on a product to view its entire journey, including origin, processing, and transportation details.
 - Pharmaceutical Supply Chain: Blockchain is employed to track pharmaceuticals throughout the supply chain, preventing counterfeiting and ensuring patient safety. Each drug's unique identifier can be linked to its blockchain record, verifying its authenticity and preventing diversion.
 - Fashion Supply Chain: Blockchain is used to track the origin and manufacturing processes of garments, promoting transparency and ethical sourcing. Consumers can gain insights into the materials used, labor practices, and environmental impact of their clothing.
- Logistics and Shipping: Blockchain is applied to optimize logistics operations, reducing delays and improving efficiency. Smart contracts can automate shipment tracking, customs clearance, and freight payments, streamlining the transportation process.

I. INTRODUCTION

In today's interconnected world, supply chain management (SCM) has become increasingly complex and challenging. Traditional SCM systems often face issues with transparency, traceability, and trust, leading to inefficiencies, fraud, and disputes. This is where blockchain technology emerges as a promising solution to revolutionize SCM practices. The potential of blockchain in SCM is not just

II. COST BENEFIT ANALYSIS

While extending payment terms can be a financing strategy for buyers, it strains suppliers' cash flow.

Supply Chain Finance (SCF) bridges this gap. It offers a solution that benefits both parties by improving working capital. Suppliers receive faster payments, and buyers can still leverage extended payment terms.

We know that digitalization has disrupted the whole industry and transformed them completely.

Blockchain has shown incredible potential in transforming the existing financial institutions and it also has the ability to increase the efficiency of transactions while being more secure and transparent.

This report gives an insight of the cost benefit analysis of blockchain based supply chain and to conduct this analysis a cost benefit model is applied to calculate the net operating processes of traditional and blockchain based supply chain managements

We are going to analyse the operating processes in three different layouts:

1. Traditional scm without the use of blockchain
2. Blockchain based scm using smart contracts
3. Blockchain based scm using iot devices and blockchain system

We then compared these three layouts to a real world scenario. The model that we use is proposed by Dello Iacono, Reindorp, Dellaert.

Nonetheless after various research it is found that this model isn't that much accurate instead it provides a base estimation of the benefits since in actual businesses there are different parameters to consider which the model is unable to cover. Also the service fee charged for using the platform is not included in this model. This is valid in only one case where the funder is the one who developed the supply chain management system.

The case study used in this analysis is real data from some of the biggest banks, which includes the operating processes and financial arrangements of SCF. The supplier in this case study supplies various raw materials, foods. Now in order to accurately calculate the benefits of SCM we put the same data in all of the three layouts. There is a set of fixed parameters which include

- Amount of the invoice
- Suppliers rate
- Buyers rate
- Original payment term
- Extended payment term

- Interest rate of SCF

After applying these parameters, the three parameters we found four key results. Blockchain technology can increase the total net benefit of SCF solutions for all involved parties (suppliers, buyers, funders, and platform providers). This is because blockchain technology can improve the efficiency of invoice processing.

The allocation of benefits changes with blockchain technology. Suppliers and buyers see a decrease in their benefit allocation, while funders and platform providers see an increase.

Suppliers are the only party whose net benefit might be negative. This is because the platform fee they pay might outweigh the benefit they earn in terms of unlocked working capital. The break-even platform fee is calculated for each scenario.

Buyers do not benefit from unlocked working capital through faster processes with blockchain technology, because the benefit to the buyer depends on the extended due date for the buyer to pay the invoice.

III. RESULTS

In view of the supplies point of view they would be able to enjoy the benefit. With a clear and immutable record of transactions, disputes over deliveries, payments, or product quality can be resolved more quickly and efficiently. This reduces friction and fosters better collaboration within the supply chain network. Blockchain can open up new opportunities for suppliers. For example, it can be used to create tamper-proof certificates for products, which can be valuable for high-value goods or those susceptible to counterfeiting.

In view of the buyer's point of view they would be able to gain a clear view of the entire supply chain, from raw materials to finished products even though they would not be able to directly benefit from the capital saved from implementing blockchain technology. This allows them to verify the origin, quality, and ethical sourcing of the goods they purchase. This can be crucial for industries with strict regulations or where counterfeiting is a concern. Also there would be greater transparency which could foster trust and collaboration between buyers and suppliers. Blockchain promotes clear communication and efficient dispute resolution, leading to stronger partnerships within the supply chain network.

The thesis concludes that understanding the costs and benefits of implementing blockchain-based SCF solutions is crucial for businesses. The cost-benefit

model presented in the thesis can be used to quantify the net value of implementing blockchain-based SCF solutions. The results suggest that blockchain technology can increase the total net benefit of involved parties in SCF as a result of improved efficiency of invoice processing.

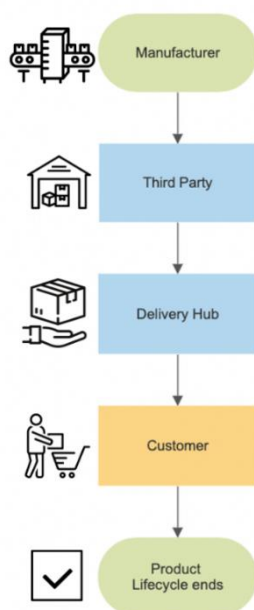
IV. BLUEPRINT

Supply chain is always hard to manage and requires a lot of administrative machinery. However, when managed with smart contracts using blockchain, a lot of the paperwork is reduced. Also it leads to an increase in transparency and helps to build an efficient Root of Trust. Supply-chain-dapp is such an implementation of a supply chain management system which uses blockchain to ensure a transparent and secure transfer of product from the manufacturer to the customer, warehouses, factories.

The smart contract will be written in solidity which is then compiled and deployed using truffle.js on the local blockchain network using Ganache-cli. The frontend uses Web3.js to communicate with the smart contract and also the local blockchain network. The requests from users are forwarded to frontend through Nginx (load balancer) and Express.js for dynamic routing.

Additionally, IoT devices can also be used which could further ease the whole process. IoT devices are also used for asset management using NFC technology and RFID tags. Overall, IoT devices are used in almost every stage of the supply chain process.

Supply Chain Flow



Solidity

Solidity is an object-oriented, high-level language for implementing smart contracts. Smart contracts are programs which govern the behavior of accounts within the Ethereum state.

React

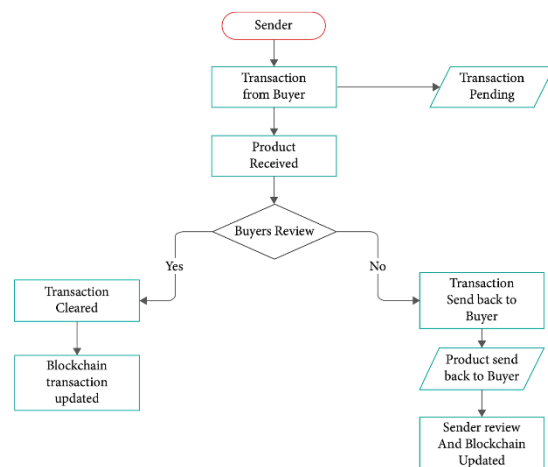
React makes it painless to create interactive UIs. It helps in Designing simple views for each state in the application, and React will efficiently update and render just the right components when the data changes.

Truffle

A world class development environment, testing framework and asset pipeline for blockchains using the Ethereum Virtual Machine (EVM), aiming to make life as a developer easier.

Ganache

It is a personal blockchain for rapid Ethereum and Filecoin distributed application development. It can be used by Ganache across the entire development cycle; enabling one to develop, deploy, and test dApps in a safe and deterministic environment. Ganache comes in two flavors: a UI and CLI. Ganache UI is a desktop application supporting Ethereum and Filecoin technology.



V. CONCLUSION

Blockchain technology is poised to revolutionize SCM, transforming the way businesses manage their supply chains. By leveraging blockchain's transparency, traceability, and automation capabilities, companies can enhance efficiency, reduce costs, and build trust among supply chain partners. As blockchain adoption continues to grow, we can expect to see even more innovative applications emerge, further optimizing SCM practices and shaping the future of global trade. Supply chain management is about to change as a consequence of blockchain technology's

improvements in efficiency, trust, and transparency. Everyone can monitor things easily with a shared, unchangeable ledger, increasing traceability and decreasing counterfeiting. Better cooperation between distributors, manufacturers, and suppliers is facilitated by this, which streamlines logistics and cuts down on delays. Consumers are empowered by transparency since it enables them to confirm the ethical sources and product origins. This encourages sustainable behaviours and increases consumer trust in companies. Moreover, blockchain-enabled automated workflows and optimized procedures can drastically save administrative expenses.

Although there are still issues with scalability and interaction with current systems, **blockchain has the unquestionable potential to completely transform supply chains.** We may anticipate a future of safe, effective, and transparent international trade as technology advances and use increases.

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